COMP233 — Discrete Mathematics

Lecture 4.1

Introduction to Number Theory & Proof Methods

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From logical statements to mathematical proofs

- Propositional and predicate logic tell us what a claim means.
- Number theory gives precise definitions to reason with.
- Proof methods tell us why a claim holds for every allowed case.
- A counterexample shows exactly why a universal claim fails.

Today: definitions → quantifiers → counterexamples → direct proof.

Why number theory matters to programmers

Programs reason about integers

Loops, counters, indexing, conditions, parity tests, and divisibility all involve integer properties.

Correctness requires general reasoning

Testing several inputs is useful, but a proof gives a guarantee for all inputs satisfying its assumptions.

Examples to investigate: even × even; odd + odd; odd × odd.

What is an even integer?

Formal definition

$n \in \mathbb{Z}$ is even $\Leftrightarrow$ $n = 2k$ for some $k \in \mathbb{Z}$.

Boundary examples

$0 = 2(0)$ → even
$-8 = 2(-4)$ → even

How to use the definition

To prove an expression E is even, rewrite E = 2 × (integer). You must justify the factor is an integer.

What is an odd integer?

Formal definition

$n \in \mathbb{Z}$ is odd $\Leftrightarrow$ $n = 2k + 1$ for some $k \in \mathbb{Z}$.

Boundary examples

$1 = 2(0) + 1$
$-301 = 2(-151) + 1$

Every integer has exactly one parity

An integer is either even or odd, never both. Negative integers and zero are included.

Classify algebraic expressions using definitions

Always even

For $a,b \in \mathbb{Z}$:
$6a^2b = 2(3a^2b)$
Since $3a^2b \in \mathbb{Z}$, the result is even.

Always odd

For $a,b \in \mathbb{Z}$:
$10a + 8b + 1 = 2(5a + 4b) + 1$
Since $5a + 4b \in \mathbb{Z}$, the result is odd.

Do not merely label parity: exhibit the required integer witness.

Prime vs. composite integers

Prime

An integer $n > 1$ is prime if its only positive divisors are 1 and n.
Examples: 2, 3, 5, 7, 11.

Composite

An integer $n > 1$ is composite if it has a positive divisor other than 1 and n.
Examples: $4 = 2 \cdot 2$; $6 = 2 \cdot 3$; $9 = 3 \cdot 3$.

Special cases: 1 is neither prime nor composite; this classification is for $n > 1$.

Concept check: classification and conditions

- Is 0 even? Which integer witnesses the definition?
- Is −301 odd? Identify k in 2k + 1.
- Is 1 prime? Why does n > 1 matter?
- Is every integer prime or composite? State the correct domain.

Always state the domain before evaluating a mathematical claim.

Universal and existential claims

Universal statement

$\forall x \in D, P(x)$
“All elements of D have property P.”
To prove: a general argument.
To disprove: one counterexample.

Existential statement

$\exists x \in D, P(x)$
“At least one element has property P.”
To prove: exhibit a witness.
To disprove: show no witness exists.

Negation connects proof and disproof

Negation of “all”

$\neg(\forall x \in D, P(x)) \Leftrightarrow \exists x \in D, \neg P(x)$
To negate a universal statement, exhibit at least one failure.

Negation of “there exists”

$\neg(\exists x \in D, P(x)) \Leftrightarrow \forall x \in D, \neg P(x)$
To disprove existence, establish failure for every element.

Choose the appropriate proof strategy

∀x P(x) — Prove

Take arbitrary x ∈ D; derive P(x).

∀x P(x) — Disprove

Find x ∈ D with ¬P(x).

∃x P(x) — Prove

Find a witness x ∈ D with P(x).

∃x P(x) — Disprove

Prove ∀x ∈ D, ¬P(x).

How to disprove a universal implication

Logical target

$\forall x \in D, [P(x) \Rightarrow Q(x)]$
Find $x \in D$ for which $P(x)$ is TRUE and $Q(x)$ is FALSE.

Counterexample checklist

1. Choose a value in the domain.
2. Verify the hypothesis $P(x)$.
3. Verify the conclusion $Q(x)$ fails.
4. State why the universal claim is false.

Counterexample: equal squares

False claim

For all $a,b \in \mathbb{R}$: if $a^2 = b^2$, then $a = b$.

Construct the counterexample

Take $a = 1$ and $b = -1$.
Hypothesis TRUE: $a^2 = 1 = b^2$.
Conclusion FALSE: $1 \neq -1$.
Therefore the statement is false.

Choosing a = b would NOT disprove the implication.

Second counterexample: squaring inequalities

False claim

For all $a,b \in \mathbb{Z}$: if $a < b$, then $a^2 < b^2$.

Counterexample

Choose $a = -3$, $b = 2$.
$-3 < 2$ is true, but $9 < 4$ is false.
Squaring does not preserve order across negative and positive inputs.

Method of exhaustion: finite cases

When exhaustive checking is a proof

If the domain is finite, list all allowed cases and establish the claim for each case. No case may be omitted.

Original lecture example

Each even n from 4 through 26 is a sum of two primes:
4=2+2 6=3+3 8=3+5 10=5+5
12=5+7 14=3+11 16=5+11 18=7+11
20=7+13 22=5+17 24=5+19 26=7+19

A finite check does not establish a corresponding claim for all positive even integers.

Direct proof: the five-step structure

- Formal restatement: identify the exact claim and domain.
- Starting point: choose arbitrary elements satisfying hypotheses.
- Unpack definitions: introduce appropriate integer witnesses.
- Derive: use valid algebra and logical reasoning.
- Conclude: verify the required definition or condition.

A direct proof of P ⇒ Q assumes P and derives Q.

Worked proof: even + even = even

Theorem

For all even integers m and n, their sum m + n is even.

Proof

Let m and n be arbitrary even integers.
By definition, $m = 2k$ and $n = 2j$, for $k,j \in \mathbb{Z}$.
Therefore $m + n = 2k + 2j = 2(k + j)$.
Because $k + j \in \mathbb{Z}$, the sum is even.

The statement “$k + j \in \mathbb{Z}$” is part of the reasoning, not optional decoration.

Worked proof: odd × odd = odd

Theorem

The product of any two odd integers is odd.

Proof

Let $m = 2a + 1$ and $n = 2b + 1$, where $a,b \in \mathbb{Z}$.

$$\begin{aligned} mn &= (2a + 1)(2b + 1) \\ &= 4ab + 2a + 2b + 1 \\ &= 2(2ab + a + b) + 1. \end{aligned}$$

Since $2ab + a + b \in \mathbb{Z}$, mn is odd.

Proof discovery: sum of two odd integers

What is known?

$m = 2a + 1$ and $n = 2b + 1$
for some integers a,b.

What must be shown?

$m + n = 2k$ for some $k \in \mathbb{Z}$.

Complete the derivation

$m + n = 2a + 1 + 2b + 1 = 2(a + b + 1)$.
Since $a + b + 1 \in \mathbb{Z}$, $m + n$ is even.

Programming connection: why proof matters

Property

If x and y are even integers, then x + y is even.
The direct proof establishes this for all mathematical integer inputs.

Implementation caveat

A programming-language integer type may have finite range or overflow behavior. The mathematical guarantee applies directly when arithmetic is exact (or overflow is excluded).

Proof assumptions should match the program’s actual semantics.

Lecture synthesis: choose your evidence

To prove

Universal → arbitrary element + general reasoning.
Existential → one verified witness.

To disprove

Universal → one valid counterexample.
Existential → show every candidate fails.

Next: Rational numbers — proofs using r = a/b with integers a, b ≠ 0.

Appendix A — Core concepts

Definitions

Even: $n = 2k$. Odd: $n = 2k+1$ ($k \in \mathbb{Z}$).
Prime: $n > 1$ with positive divisors $1, n$ only.
Composite: $n > 1$ with another positive divisor.

Proof essentials

Proof must cover all required values.
A counterexample must violate the actual claim.
Final algebraic factor must belong to the required domain.

Appendix B — Connections and distinctions

Key equivalences

$\neg \forall x\, P(x) \Leftrightarrow \exists x\, \neg P(x)$
$\neg \exists x\, P(x) \Leftrightarrow \forall x\, \neg P(x)$
“Even n” $\Leftrightarrow \exists k \in \mathbb{Z}: n = 2k$
“Odd n” $\Leftrightarrow \exists k \in \mathbb{Z}: n = 2k+1$

Important distinction

Testing examples supplies evidence; proving a universally quantified statement requires justification across its entire domain.

Appendix C — Examples and boundary cases

Valid / boundary

$0 = 2(0)$ even; $-7 = 2(-4)+1$ odd.
1 is neither prime nor composite.
2 is prime and even.

Invalid universal claims

$a^2=b^2$ does not imply $a=b$ over $\mathbb{R}$.
$a<b$ does not imply $a^2<b^2$ over $\mathbb{Z}$.
Use $a=1, b=-1$; and $a=-3, b=2$ respectively.

Appendix D — Knowledge check 1/8

Q1 · Multiple choice

Which is an even integer?

A -7 B 0 C 9 D 11

Explain your reasoning; do not provide only a final answer.

Appendix D — Knowledge check 2/8

Q2 · Justified true / false

“Every integer greater than 1 is either prime or composite.”
State the relevant definition.

Explain your reasoning; do not provide only a final answer.

Appendix D — Knowledge check 3/8

Q3 · Error detection

A student writes: “$6a^2b$ is even because it contains 6.”
Repair the argument for arbitrary integers a,b.

Explain your reasoning; do not provide only a final answer.

Appendix D — Knowledge check 4/8

Q4 · Quantifiers

Write the negation of:
$\forall n \in \mathbb{Z}, [\text{Even}(n) \Rightarrow \text{Even}(n^2)]$.

Explain your reasoning; do not provide only a final answer.

Appendix D — Knowledge check 5/8

Q5 · Counterexample

Disprove: For all $a,b \in \mathbb{R}$, $a^2=b^2 \Rightarrow a=b$.
Show why your choices meet the counterexample conditions.

Explain your reasoning; do not provide only a final answer.

Appendix D — Knowledge check 6/8

Q6 · Proof construction

Prove: the sum of two odd integers is even.
State the integer witnesses explicitly.

Explain your reasoning; do not provide only a final answer.

Appendix D — Knowledge check 7/8

Q7 · Proof / disproof

Claim: The product of an even integer and any integer is even.
Prove or disprove, with justification.

Explain your reasoning; do not provide only a final answer.

Appendix D — Knowledge check 8/8

Q8 · Higher-order reasoning

A program tests “every even $n \geq 4$ is a sum of two primes” for $n \leq 1{,}000{,}000$. Has it proved the universal claim? Explain.

Explain your reasoning; do not provide only a final answer.

Appendix E — Answer key 1/8

A1 · Explanation

B. Zero is even because 0 = 2·0. Odd choices cannot be written as 2k with integer k.

Appendix E — Answer key 2/8

A2 · Explanation

True. By definition, every n > 1 has either no positive divisor beyond 1 and n (prime) or at least one (composite).

Appendix E — Answer key 3/8

A3 · Explanation

$6a^2b = 2(3a^2b)$. Since $3a^2b \in \mathbb{Z}$, the expression is even. “Contains 6” is not a formal proof.

Appendix E — Answer key 4/8

A4 · Explanation

$\exists n \in \mathbb{Z}$ such that $\text{Even}(n) \wedge \neg\text{Even}(n^2)$. The original implication is actually true; its negation has no witness.

Appendix E — Answer key 5/8

A5 · Explanation

Set $a = 1$, $b = -1$. Their squares are equal but $a \neq b$, so the implication fails for this pair.

Appendix E — Answer key 6/8

A6 · Explanation

Let $m=2a+1, n=2b+1$ for $a,b \in \mathbb{Z}$. Then $m+n=2(a+b+1)$; $a+b+1 \in \mathbb{Z}$, hence even.

Appendix E — Answer key 7/8

A7 · Explanation

True. Let $m=2k$ and $n \in \mathbb{Z}$. Then $mn=2(kn)$, and $kn \in \mathbb{Z}$; thus mn is even.

Appendix E — Answer key 8/8

A8 · Explanation

No. Finite testing proves the claim only over the checked finite range. It does not establish the statement for all even $n \geq 4$.

Appendix F — Optional enrichment

Program verification

A mathematical proof of parity uses unbounded integers; programming implementations must account for finite integer types and language semantics.

From parity to divisibility

The expression n = 2k is a special case of divisibility: 2 divides n. This idea leads directly into Lecture 4.3.

References and course resources

- Primary reference: Susanna S. Epp, Discrete Mathematics with Applications, 4th ed.
- Original course materials: Prof. Mustafa Jarrar — www.jarrar.info/courses/DMath/
- Updated course resources: github.com/RuwaYafa/BZU_Courses/tree/main/COMP233
- Original lecture materials also circulated with minor contributions from Dr. Radi Jarrar.

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